```
        image: mennayasser777/ivolve-app:v1
        ports:
          - containerPort: 3000
mennatallahyasser@master:~$ kubectl apply -f deployment.yaml
deployment.apps/ivolveapp created
mennatallahyasser@master:~$ kubectl get deployment,pods -n ivolve -o wide
NAME                        READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES                          SELECTOR
deployment.apps/ivolveapp   0/2     2            0           9s    ivolveapp    mennayasser777/ivolve-app:v1   app=ivolveApp

NAME                                READY   STATUS    RESTARTS       AGE     IP              NODE      NOMINATED NODE   READINESS GATES
pod/ivolve-db-0                     1/1     Running   1 (4h2m ago)   3d18h   10.244.235.130   worker1   <none>           <none>
pod/ivolveapp-5c5ff5c588-fhdmg      0/1     Pending   0              9s      <none>          <none>    <none>           <none>
pod/ivolveapp-5c5ff5c588-kj6zh      0/1     Pending   0              9s      <none>          <none>    <none>           <none>
mennatallahyasser@master:~$ nano deployment.yaml
mennatallahyasser@master:~$ kubectl apply -f deployment.yaml
deployment.apps/ivolveapp configured
mennatallahyasser@master:~$ kubectl get pods -n ivolve -o wide
NAME                        READY   STATUS    RESTARTS      AGE     IP              NODE      NOMINATED NODE   READINESS GATES
ivolve-db-0                 1/1     Running   1 (10h ago)   4d1h    10.244.235.130   worker1   <none>           <none>
ivolveapp-5c5ff5c588-fhdmg  0/1     Pending   0             6h29m   <none>          <none>    <none>           <none>
ivolveapp-5c5ff5c588-kj6zh  0/1     Pending   0             6h29m   <none>          <none>    <none>           <none>
mennatallahyasser@master:~$ nano deployment.yaml
mennatallahyasser@master:~$ kubectl apply -f deployment.yaml
deployment.apps/ivolveapp configured
mennatallahyasser@master:~$ kubectl get pods -n ivolve -o wide -w
NAME                        READY   STATUS              RESTARTS      AGE     IP              NODE      NOMINATED NODE   READINESS GATES
ivolve-db-0                 1/1     Running             1 (10h ago)   4d1h    10.244.235.130   worker1   <none>           <none>
ivolveapp-5c5ff5c588-kj6zh  0/1     Pending             0             6h41m   <none>          <none>    <none>           <none>
ivolveapp-7d4bbd5857-jpws9  0/1     ContainerCreating   0             6s      <none>          worker1   <none>           <none>
ivolveapp-7d4bbd5857-jpws9  1/1     Running             0             12s     10.244.235.131   worker1   <none>           <none>
ivolveapp-5c5ff5c588-kj6zh  0/1     Terminating         0             6h41m   <none>          <none>    <none>           <none>
ivolveapp-5c5ff5c588-kj6zh  0/1     Terminating         0             6h41m   <none>          <none>    <none>           <none>
ivolveapp-7d4bbd5857-2tpm5  0/1     Pending             0             0s      <none>          <none>    <none>           <none>
ivolveapp-7d4bbd5857-2tpm5  0/1     Pending             0             0s      <none>          worker1   <none>           <none>
ivolveapp-7d4bbd5857-2tpm5  0/1     ContainerCreating   0             0s      <none>          worker1   <none>           <none>
ivolveapp-7d4bbd5857-2tpm5  0/1     ContainerCreating   0             1s      <none>          worker1   <none>           <none>
ivolveapp-7d4bbd5857-2tpm5  1/1     Running             0             2s      10.244.235.132   worker1   <none>           <none>
mennatallahyasser@master:~$
```

master | My Computer | worker1

GNU nano 8.1 — deployment.yaml — Modified

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ivolveapp
  namespace: ivolve
  labels:
    app: ivolveApp
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ivolveApp
  template:
    metadata:
      labels:
        app: ivolveApp
    spec:
      nodeSelector:
        project: ivolve
      tolerations:
        - key: workload
          operator: Equal
          value: ivolveApp
          effect: NoSchedule
      volumes:
        - name: app-logs
          persistentVolumeClaim:
            claimName: ivolve-pvc
      initContainers:
        - name: init-db
          image: mysql:8.0
          envFrom:
            - configMapRef:
                name: ivolve-cm
            - secretRef:
                name: ivolve-secret
          command:
            - sh
            - -c
            - |
              until mysqladmin ping -h "$DB_HOST" -uroot -p"$MYSQL_ROOT_PASSWORD" --silent; do
                echo "Waiting for MySQL..."; sleep 3
              done
              mysql -h "$DB_HOST" -uroot -p"$MYSQL_ROOT_PASSWORD" -e "
                CREATE DATABASE IF NOT EXISTS ivolve;
                CREATE USER IF NOT EXISTS '$DB_USER'@'%' IDENTIFIED BY '$DB_PASSWORD';
                GRANT ALL PRIVILEGES ON ivolve.* TO '$DB_USER'@'%';
                FLUSH PRIVILEGES;"
              echo "Database ivolve is ready"
      containers:
        - name: ivolveapp
          image: mennayasser777/ivolve-app:v1
          ports:
            - containerPort: 3000
          envFrom:
            - configMapRef:
                name: ivolve-cm
            - secretRef:
                name: ivolve-secret
          env:
            - name: DB_NAME
              value: ivolve
          volumeMounts:
            - name: app-logs
              mountPath: /app/logs
```

```
ivolveapp-7d4bbd5857-jpws9   1/1     Running     0          20m

^Cmernatallahyasser@master:~$ kubectl logs -n ivolve deployment/ivolveapp -c init-db
Found 2 pods, using pod/ivolveapp-7d4bbd5857-jpws9
error: container init-db is not valid for pod ivolveapp-7d4bbd5857-jpws9
mernatallahyasser@master:~$ kubectl logs -n ivolve deployment/ivolveapp -c init-db
error: container init-db is not valid for pod ivolveapp-7d4bbd5857-jpws9
mernatallahyasser@master:~$ kubectl get pods -n ivolve -w
NAME                         READY   STATUS    RESTARTS      AGE
ivolve-db-0                  1/1     Running   1 (11h ago)   4d1h
ivolveapp-7d4bbd5857-jpws9   1/1     Running   0             31m
ivolveapp-76955764c9-vmg4k   0/1     Pending   0             0s
ivolveapp-76955764c9-vmg4k   0/1     Pending   0             0s
ivolveapp-76955764c9-vmg4k   0/1     Init:0/1  0             0s
ivolveapp-76955764c9-vmg4k   0/1     Init:0/1  0             1s
ivolveapp-76955764c9-vmg4k   0/1     Init:0/1  0             3s
ivolveapp-76955764c9-vmg4k   0/1     PodInitializing   0           4s
ivolveapp-76955764c9-vmg4k   1/1     Running           0           5s
ivolveapp-7d4bbd5857-jpws9   1/1     Terminating       0           33m
ivolveapp-7d4bbd5857-jpws9   1/1     Terminating       0           33m
ivolveapp-7d4bbd5857-jpws9   1/1     Terminating       0           33m
ivolveapp-7d4bbd5857-jpws9   0/1     Error             0           33m
ivolveapp-7d4bbd5857-jpws9   0/1     Error             0           33m
ivolveapp-7d4bbd5857-jpws9   0/1     Error             0           33m
ivolveapp-76955764c9-784p2   0/1     Pending           0           0s
ivolveapp-76955764c9-784p2   0/1     Pending           0           0s
ivolveapp-76955764c9-784p2   0/1     Init:0/1          0           0s
ivolveapp-76955764c9-784p2   0/1     Init:0/1          0           1s
ivolveapp-76955764c9-784p2   0/1     Init:0/1          0           1s
ivolveapp-76955764c9-784p2   0/1     PodInitializing   0           2s
ivolveapp-76955764c9-784p2   1/1     Running           0           3s
```

```
  GNU nano 8.1                                        service.yaml
apiVersion: v1
kind: Service
metadata:
  name: ivolveappsvc
  namespace: ivolve
spec:
  type: ClusterIP
  selector:
    app: ivolveApp
  ports:
    - port: 80
      targetPort: 3000
```

```
mennatallahyasser@master:~$ nano service.yaml
mennatallahyasser@master:~$ kubectl apply -f service.yaml
service/ivolveappsvc created
mennatallahyasser@master:~$ kubectl get svc -n ivolve
NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)    AGE
ivolve-db-svc   ClusterIP   None           <none>        3306/TCP   4d2h
ivolveappsvc    ClusterIP   10.108.171.5   <none>        80/TCP     4s
mennatallahyasser@master:~$ kubectl get endpoints ivolveappsvc -n ivolve
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME           ENDPOINTS                                   AGE
ivolveappsvc   10.244.235.133:3000,10.244.235.134:3000   13s
mennatallahyasser@master:~$
```

```
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
command terminated with exit code 1
mennatallahyasser@master:~$ kubectl exec -it ivolve-db-0 -n ivolve -- mysql -h ivolve-db-svc -u root -pRootPass123
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 785
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> ^C
mysql>
```

```
mysql> ^C
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| ivolve             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.16 sec)

mysql>
```

```
| sys                |
+--------------------+
5 rows in set (0.16 sec)

mysql> SELECT user, host FROM mysql.user;
+------------------+-----------+
| user             | host      |
+------------------+-----------+
| ivolve_user      | %         |
| root             | %         |
| mysql.infoschema | localhost |
| mysql.session    | localhost |
| mysql.sys        | localhost |
| root             | localhost |
+------------------+-----------+
6 rows in set (0.01 sec)

mysql>
```

```
6 rows in set (0.01 sec)

mysql> SHOW GRANTS FOR 'ivolve_user'@'%';
+-----------------------------------------------------------+
| Grants for ivolve_user@%                                  |
+-----------------------------------------------------------+
| GRANT USAGE ON *.* TO `ivolve_user`@`%`                   |
| GRANT ALL PRIVILEGES ON `ivolve`.* TO `ivolve_user`@`%`   |
+-----------------------------------------------------------+
2 rows in set (0.00 sec)

mysql>
```

```
mennatallahyasser@master:~$ kubectl logs -n ivolve -l app=ivolveApp -c ivolveapp
🔄 Attempting to connect to MySQL...
✅ Connected to MySQL and 'ivolve' DB found.
🚀 Server started on http://0.0.0.0:3000
🔄 Attempting to connect to MySQL...
✅ Connected to MySQL and 'ivolve' DB found.
🚀 Server started on http://0.0.0.0:3000
mennatallahyasser@master:~$
```